# EXPERIMENT - 2

**AIM:** To Implement Multi Regression, Lasso, and Ridge Regression on real-world datasets.

**OUTPUT:**

## Loading the dataset

```
   age         job  marital  education default  balance housing loan  contact
0   59      admin.  married  secondary      no     2343     yes   no  unknown
1   56      admin.  married  secondary      no       45      no   no  unknown
2   41  technician  married  secondary      no     1270     yes   no  unknown
3   55    services  married  secondary      no     2476     yes   no  unknown
4   54      admin.  married   tertiary      no      184      no   no  unknown

   day month  duration  campaign  pdays  previous poutcome deposit
0    5   may      1042         1     -1         0  unknown     yes
1    5   may      1467         1     -1         0  unknown     yes
2    5   may      1389         1     -1         0  unknown     yes
3    5   may       579         1     -1         0  unknown     yes
4    5   may       673         2     -1         0  unknown     yes

Shape: (11162, 17)

Missing Values:
 age          0
job          0
marital      0
education    0
default      0
balance      0
housing      0
loan         0
contact      0
day          0
month        0
duration     0
campaign     0
pdays        0
previous     0
poutcome     0
deposit      0
dtype: int64
```

## Encode Categorical Variables

```
   age  balance  day  duration  campaign  pdays  previous  job_blue-collar
0   59     2343    5      1042         1     -1         0            False
1   56       45    5      1467         1     -1         0            False
2   41     1270    5      1389         1     -1         0            False
3   55     2476    5       579         1     -1         0            False
4   54      184    5       673         2     -1         0            False

   job_entrepreneur  job_housemaid  ...  month_jun  month_mar  month_may  \
0             False          False  ...      False      False       True
1             False          False  ...      False      False       True
2             False          False  ...      False      False       True
3             False          False  ...      False      False       True
4             False          False  ...      False      False       True

   month_nov  month_oct  month_sep  poutcome_other  poutcome_success  \
0      False      False      False           False             False
1      False      False      False           False             False
2      False      False      False           False             False
3      False      False      False           False             False
4      False      False      False           False             False

   poutcome_unknown  deposit_yes
0              True         True
1              True         True
2              True         True
3              True         True
4              True         True
```

## Multiples linear regression results

```
Linear Regression Results
MSE: 10115403.096451918
R2 Score: 0.04066807071426137
```

## Ridge regression results

```
Ridge Regression Results
MSE: 10115380.582195144
R2 Score: 0.04067020593766091
```

## Lasso regression results

```
Lasso Regression Results
MSE: 10115169.963667862
R2 Score: 0.04069018073018216
```

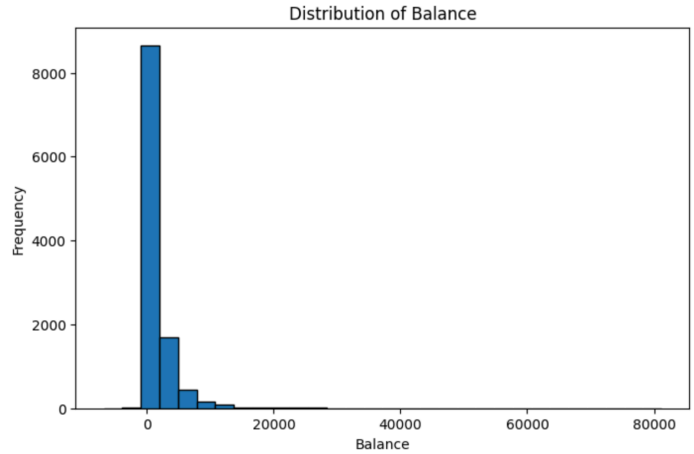
## Hyperparameter tuning (Ridge regression)

```
Best Ridge Alpha: {'alpha': 100}
Best Ridge Score: 0.03317293126309688
```

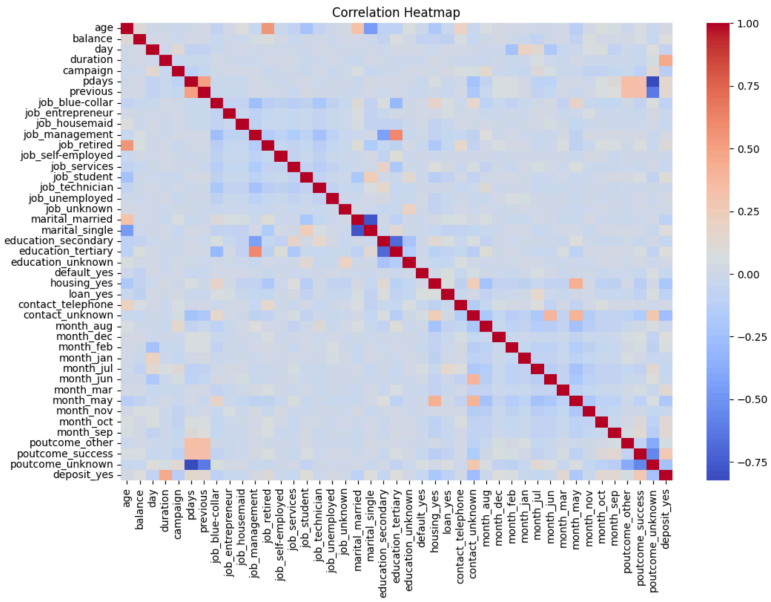
## Hyperparameter tuning (LAsso regression)

```
Best Lasso Alpha: {'alpha': 10}
Best Lasso Score: 0.03419675564705109
```

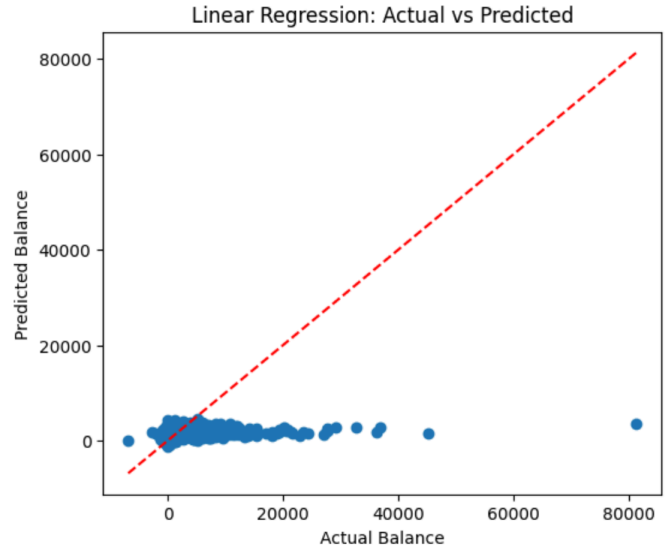
## Histogram of Target Variable (Balance)



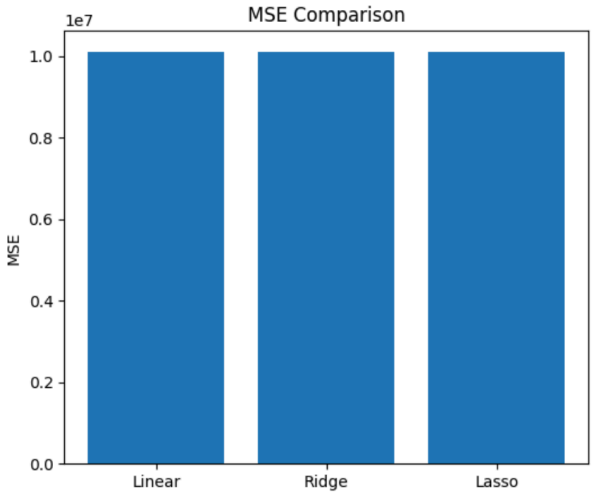
## Correlation Heatmap



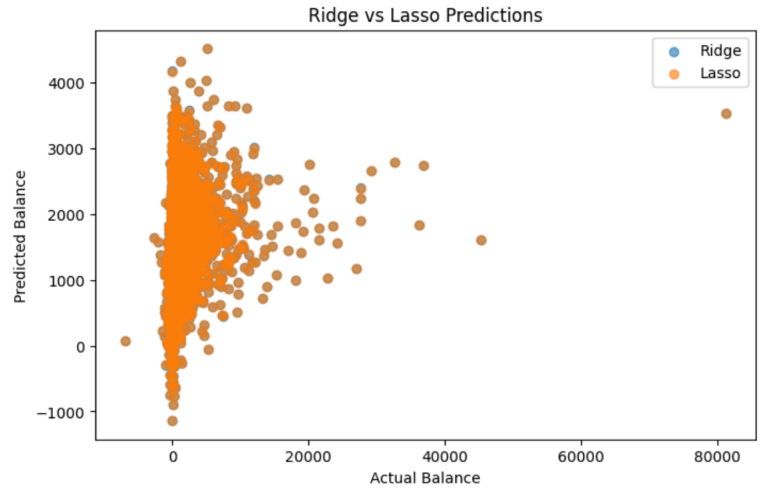

## Actual vs Predicted (Linear Regression)

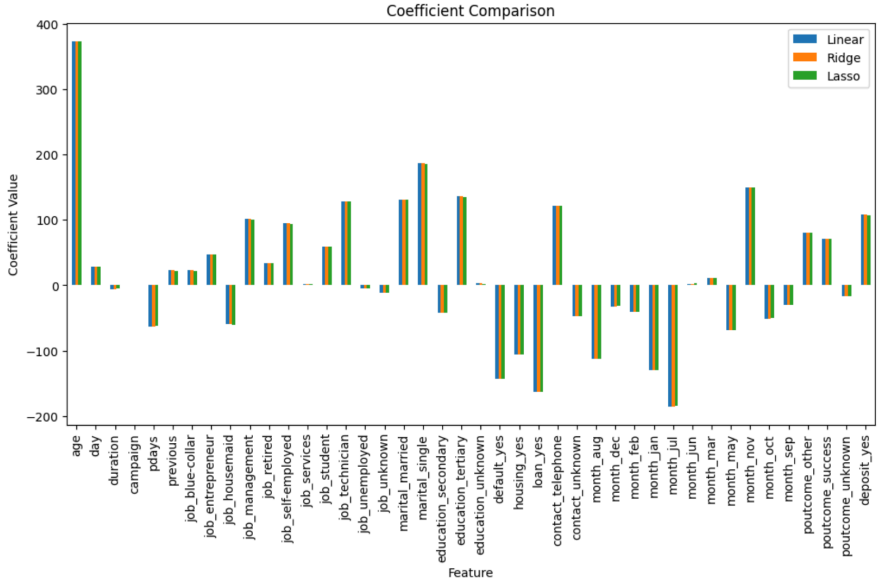



## MSE Comparison Bar Graph




## Ridge vs Lasso Predictions Comparison




## Compare Coefficients (Linear vs Ridge vs Lasso) bar plot